

A Computational Model of Reactive Oxygen Species and Redox Balance in Cardiac Mitochondria

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ABSTRACT Elevated levels of reactive oxygen species (ROS) play a critical role in cardiac myocyte signaling in both healthy and diseased cells. Mitochondria represent the predominant cellular source of ROS, specifically the activity of complexes I and III. The model presented here explores the modulation of electron transport chain ROS production for state 3 and state 4 respiration and the role of substrates and respiratory inhibitors. Model simulations show that ROS production from complex III increases exponentially with membrane potential ($\Delta\Psi_m$) when in state 4. Complex I ROS release in the model can occur in the presence of NADH and succinate (reverse electron flow), leading to a highly reduced ubiquinone pool, displaying the highest ROS production flux in state 4. In the presence of ample ROS scavenging, total ROS production is moderate in state 3 and increases substantially under state 4 conditions. The ROS production model was extended by combining it with a minimal model of ROS scavenging. When the mitochondrial redox status was oxidized by increasing the proton permeability of the inner mitochondrial membrane, simulations with the combined model show that ROS levels initially decline as production drops off with decreasing $\Delta\Psi_m$ and then increase as scavenging capacity is exhausted. Hence, this mechanistic model of ROS production demonstrates how ROS levels are controlled by mitochondrial redox balance.

INTRODUCTION

Experimental evidence suggests that low levels of reactive oxygen species (ROS) play a critical role in the cellular signaling pathways of healthy cardiac myocytes (1), including the protective effects of ischemic preconditioning (2). However, ROS contribute to pathophysiology at higher concentrations. Oxidative stress can sensitize the mitochondria to permeability transition pore opening (3) and can also modulate the activity of many key ion channels and excitation-contraction coupling proteins in the cardiac myocyte. High levels of ROS contribute to contractile dysfunction and fatal arrhythmias in diseased hearts (4) and have been implicated in the progression of heart failure (5–7), particularly ROS derived from the mitochondrial electron transport chain (ETC) (8).

Cellular ROS levels depend not only on the rate of ROS production, but on the rate of ROS scavenging. The redox-optimized ROS balance (R-ORB) hypothesis (9) posits that ROS levels are minimized in an intermediate cellular redox environment, neither too reduced nor too oxidized, when energy output is maximal, but increase when the system deviates toward a more oxidized or more reduced state. This property arises from the dual control of ROS production and scavenging, respectively, by NADH redox potential, which determines the electron transport rate, and the glutathione (GSH) and nicotinamide adenine dinucleotide phosphate (NADPH) redox potentials, which drive antioxidant enzyme reactions. As the redox environment becomes more reduced than that correspond-

ing to minimal ROS levels, ROS scavenging rates are maintained, but higher mitochondrial NADH concentrations lead to a high level of reduction of the ETC complexes, which increases ROS production above the rate of scavenging. On the other hand, as the redox environment becomes more oxidized than that corresponding to minimal ROS levels, mitochondrial ROS production is low, but decreased availability of reducing equivalents for ROS scavenging can also result in accumulation of cellular ROS. ROS balance is maintained in an intermediate redox environment that is reduced enough to support high scavenging rates but oxidized enough to prevent high ROS production.

In ventricular myocytes isolated from failing hearts, oxidation of NAD(P)H occurs during increased workload (10), leading to increased cellular ROS accumulation (11), likely stemming from decreased scavenging capacity in an increasingly oxidized redox environment. In contrast, hypoxia, which reduces the cellular redox environment, can also lead to increased ROS and activation of hypoxia-inducible factor-dependent signaling (12). Furthermore, experiments in isolated mitochondria have shown that ROS production varies with NADH redox potential (13–15), the concentration (16) and type of respiratory substrate used (9,15), mitochondrial membrane potential ($\Delta\Psi_m$) (17,18), and direction of electron transport through complex I (9,15). Understanding how mitochondrial ROS production varies under different metabolic conditions will help elucidate how redox homeostasis is maintained in healthy cells and how metabolic changes in diseases like heart failure can lead to pathological levels of ROS production.

NADH-ubiquinone oxidoreductase (complex I of the ETC), is one proposed source of mitochondrial ROS

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production (13–16,19–22). Its role in respiration is to transfer electrons from NADH, a product of the tricarboxylic acid (TCA) cycle or β -oxidation, to ubiquinone (Q). Complex I is a large protein, consisting of up to 45 subunits (23) depending on species. Its structure was not fully resolved until recently (24), and its mechanism of proton pumping is still a source of controversy (25,26). A variety of conditions have been shown to elicit ROS production from complex I, including rotenone inhibition in the presence of NADH (13–15,19,20) and reverse electron transport (RET) in the presence of NADH, succinate, and high $\Delta\Psi_m$ (15,16,19,21,22). However, the source of the electrons forming superoxide is still under debate (27). It has been identified by some (14,15) as the flavin mononucleotide (FMN) cluster in the hydrophilic arm of the protein and by others (22,28) as the Q-site at the interface of the protein's matrix and membrane domains.

Mitochondrial ubiquinol-cytochrome-*c* oxidoreductase (complex III) transfers electrons from ubiquinol to cytochrome *c* by means of the Q cycle. By virtue of the fact that the high- and low-potential b-hemes (b_L and b_H) of complex III are situated in the membrane, electron transfers to and from those centers are dependent on the electric field imposed by $\Delta\Psi_m$. As $\Delta\Psi_m$ increases, the rates of electron transfer through the b-hemes decrease and the concentration of the upstream ubisemiquinone on the cytoplasm-facing side increases. One proposed mechanism for $\Delta\Psi_m$ -dependent ROS production from complex III attributes superoxide formation to the escape of an electron from the unstable ubisemiquinone to reduce oxygen. Experimental data have

confirmed the existence of a semiquinone on the cytoplasm-facing side (29,30), suggesting a mechanism for ROS production by complex III (20,31,32). Additional experiments have further elaborated on this mechanism by proposing complex III ROS release to both sides of the IMM (20,31). In contrast to complex I, high ROS production from complex III occurs during forward electron transport (FET).

In this work, we formulate a biophysically derived computational model of the ETC to investigate how changes in the mitochondrial environment control the rate of ROS production. This model is constructed with a minimal number of states to facilitate its future incorporation into multi-scale models of mitochondria and cardiac myocytes. We extend this model to include a previously published minimal model of mitochondrial ROS scavenging (33) to show how, as the redox environment varies from highly reduced to highly oxidized, opposite but complementary changes in the rate of ROS production and the ROS scavenging capacity produce a minimal level of mitochondrial ROS at intermediate mitochondrial redox potentials, in agreement with the R-ORB hypothesis (9).

METHODS

Model description

Fig. 1 A schematically depicts protein structures and redox centers of the respiratory complexes represented in this electron transport chain model. Briefly, forward and reverse rate constants for electron-transfer models of complexes II–IV were modified from Demin et al. (34,35) (see Fig. 1, C

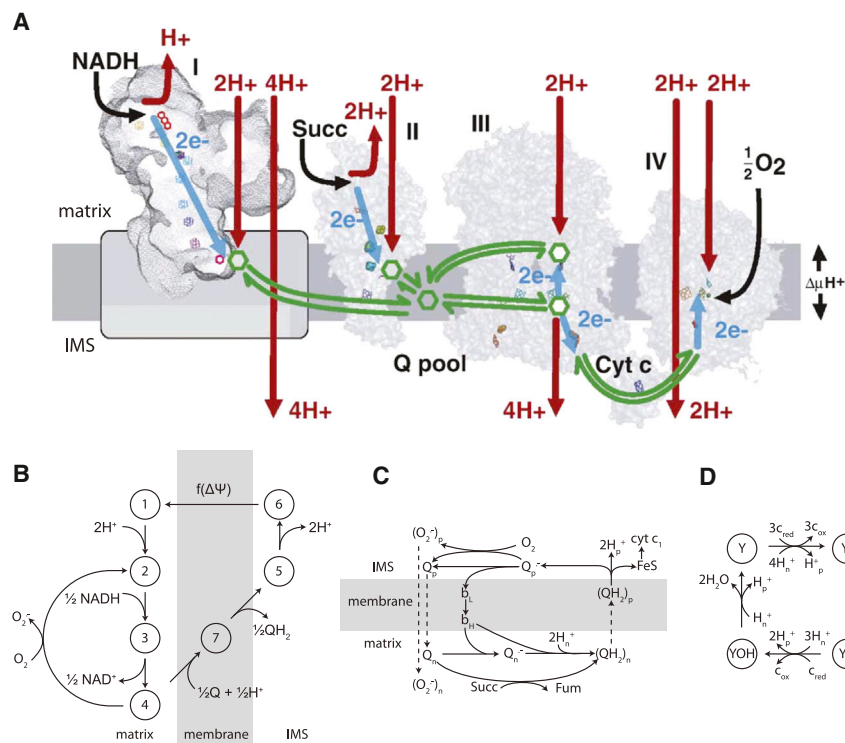


FIGURE 1 Overview of the mitochondrial ETC and model representation. (A) Mitochondrial respiratory complexes I–IV from Moser et al. ((81), reproduced with permission from Elsevier). (B) State diagram for the complex I model, modified from Magnus and Keizer (37). (C) Reaction scheme for complexes II and III, from Demin et al. (35). (D) State diagram describing the reactions of complex IV, from Demin et al. (34). Matrix and intermembrane space (IMS) compartments are indicated for topologically explicit diagrams.

and *D*) to better match data on the redox state of the ETC from Kim et al. (36). A thermodynamic model of complex I was constructed (see Fig. 1 *B*) based on the whole ETC model of Magnus and Keizer (37), and its rate constants were parameterized to match respiration and ROS production data from Aon et al. (9). Enzyme concentrations and thermodynamic quantities remained unchanged with the exception of complex I protein concentration. Overall, 23 parameters were obtained from experimental results or previous models and 39 parameters were adjusted based on experimental data. More complete methods, equations, and parameters can be found in the [Supporting Material](#).

RESULTS

In the following sections, we demonstrate the ability of the model to simulate data from which model parameters were estimated, as well as to predict experimental data not included in the optimization process.

Model reconstructions

Redox state of the ETC

ROS can only be produced from reduced redox centers, since those centers act as electron donors to reduce oxygen to superoxide. As such, the oxidation state of the redox components of the ETC can offer important information about likely sites of ROS production and how those sites are modulated by changes in the mitochondrial environment. The ETC model presented here includes variables describing the concentrations of ubiquinone, ubiquinol, and ubiquinol, along with the oxidation states of cytochrome *c* and the redox centers in complex III, i.e., the high- and low-potential b-hemes (b_H and b_L) and cytochrome c_1 . Modeling results were obtained for a range of $\Delta\Psi_m$, from the high $\Delta\Psi_m$ associated with state 4 respiration, through the state 3 $\Delta\Psi_m$ range to lower values of $\Delta\Psi_m$ found after the addition of protonophoric uncouplers. These computational results are compared with experimental data from the literature in Fig. 2, A–C. Fig. 2 *A* shows the reduction in the ubiquinone redox potential that occurs at higher $\Delta\Psi_m$ as the concentration of ubiquinone (Q) falls and ubiquinol (QH₂) increases. It is important to note that different tissue types show different mitochondrial NAD⁺/NADH redox potentials at rest. Estimations using ratios of other metabolites indicate that rat heart maintains a lower NAD⁺/NADH ratio from 1 to 16 (38,39), liver attains ratios from 30 to 100 (38), and alveolar macrophages have a ratio near 30 (40). The agreement of the redox potential of these upstream electron carriers in Fig. 2 *A* helps to validate the oxidation-state results of the downstream electron carriers. Fig. 2 *B* shows the typical $\Delta\Psi_m$ -dependent reduction of the b-hemes. This is expected, because these redox centers are embedded in the membrane, and electron transfers from the cytosolic to the matrix face of the membrane are subject to the opposing force of the membrane potential. The increased reduction of b_L plays a critical role in the mechanism of complex III ROS production in the model

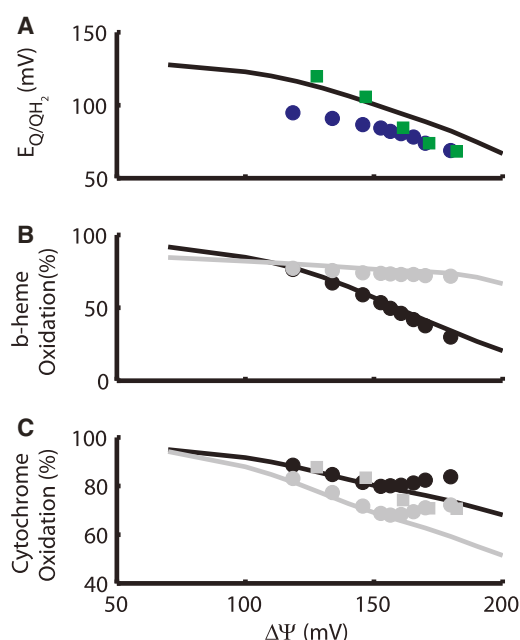


FIGURE 2 Electron carrier redox state versus mitochondrial membrane potential ($\Delta\Psi_m$). (A) Model results for redox potential of Q/QH₂ (black line) follow a decrease in redox potential with increasing $\Delta\Psi_m$ similar to the experimental data of Kim et al. (36) (circles) and Brown and Brand (41) (squares). (B) Model results (lines) are in good agreement with the increasing reduction of the b_L (black) and b_H (gray) redox centers shown by experimental measurements from Kim et al. (36) (circles). (C) Model results (lines) show increasing reduction of cytochrome c_1 (black) and c (gray), similar to the trend in cytochrome c redox state shown by Brown and Brand (41) (squares). Experiments of Kim et al. (36) (circles) show a slight reoxidation at potentials exceeding 160 mV.

presented here, in which the buildup of electrons on the cytosolic side of the Q cycle causes an increase in the ubiquinol concentration and subsequent release of its unpaired electron to form superoxide. In the model, superoxide resulting from this process is released to either the matrix or the intermembrane space with equal probability, according to the mechanisms proposed by Muller et al. (31). Fig. 2 *C* validates the $\Delta\Psi_m$ dependence of the redox states of cytochromes c_1 and c , which play a role in the transfer of electrons between complex III and complex IV. The agreement of model results and experimental data from Brown and Brand (41) in isolated mitochondria is reasonably good over the range of physiological $\Delta\Psi_m$ (120–180 mV). In contrast, data from Kim et al. (36) show that in whole macrophage cells, oxidation of cytochrome c_1 and c increases with $\Delta\Psi$ over the range 150–180 mV. This may be related to the high [ATP]/[ADP]-ratio-induced inhibition of complex IV (42), which is common in cells but not in isolated mitochondria (43). Given that the model presented here represents isolated mitochondria, the disparity between the model results and the experimental data from cells may be due to a control mechanism that is not represented in the model. It is also important to note that studies using metabolic control analysis, such as those by Rossignol et al.

(44), have revealed that metabolic control can vary widely between tissue types, which could potentially explain the difference between the Brown and Brand (41) liver data and the Kim et al. (36) macrophage data.

Control of respiration

See the [Supporting Material](#) for results validating the dependence of respiratory flux on $\Delta\Psi$ and NADH.

Control of ROS production

When complex I rates are fitted to guinea pig cardiac mitochondrial data from Aon et al. (9), the model reproduces the experimental patterns of ROS production in the presence of glutamate/malate or succinate in states 3 and 4 (Fig. 3). ROS production in complex III exhibits higher rates for state 4 than for state 3. The conditions for the highest levels of total ROS production correspond to state 4 respiration in the presence of complex II substrates. For the RET conditions in state 4 in the presence of succinate, 72% of ROS production is derived from complex I, though the contribution from the ubisemiquinone site of complex III is significant at 28%. After the addition of ADP, ROS production by both complexes, I and III, decreases substantially. For complex I substrates, complex I makes a minimal contribution to ROS production in state 3. Given that mitochondrial succinate levels have been measured at 0.3 mM (45), much lower than the 4–10 mM used to induce RET (9,15,22) in isolated mitochondria, and that complex I substrates are always present in cells, it is unlikely that physiological conditions

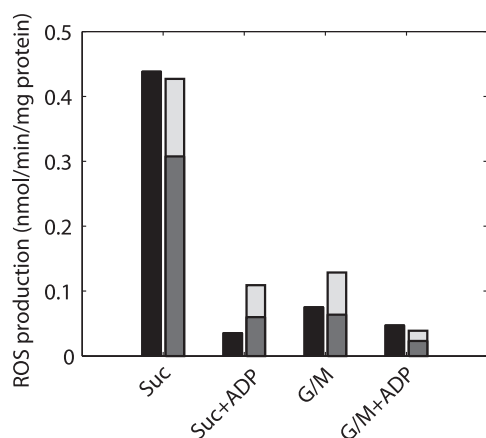


FIGURE 3 ROS production as a function of different substrates and mitochondrial respiratory states. Simulated state 4 ROS production in succinate (Suc; dark gray/light gray bar), predominated by RET-induced complex I ROS, quantitatively matches the experimental data (black). The bottom segment of each bar (dark gray) for the simulations depicts the ROS contribution from complex I, whereas the upper segment (light gray) depicts the contribution from complex III ROS production. During maximal rates of oxidative phosphorylation, stimulated by ADP (state 3), ROS production with succinate is minimized (Suc+ADP). ROS production with the complex I substrates (G/M) is split between complex I and complex III and is diminished when oxidative phosphorylation flux is maximal (G/M+ADP).

favor RET in vivo under nonischemic conditions (46). This suggests that FET ROS production would predominate physiologically. In the absence of pathological conditions leading to complex I inhibition, model simulations therefore support complex III as the primary contributor to superoxide production in normal cells.

Model predictions

After the model parameters were adjusted to reproduce the experimental data featured previously in the Model reconstructions section, the final parameter set was used to simulate additional experimental protocols. The model results using these protocols are able to reproduce data that were not included in the model parameterization process.

Control of ROS production

For mitochondria respiring on succinate in state 4, ROS production has been shown to be highly dependent on $\Delta\Psi_m$. Fig. 4 compares the model prediction of the normalized rate of ROS production from complex III as a function of $\Delta\Psi_m$ to experimental data from rat heart mitochondria (18) and isolated complex III (17). In the model, this property arises from the reduction in the rate of Q-cycle reactions involving the b-hemes as membrane potential increases. As the b-hemes become more reduced, the ability to accept electrons from ubisemiquinone on the side of the membrane facing the cytoplasm decreases, leading to a buildup of that species. Electrons from the ubisemiquinone are transferred to oxygen to produce superoxide. It is important to note that the relationship between respiratory rate and ROS production shown in Fig. 4 is derived by modulating $\Delta\Psi_m$ through increased membrane proton leak. Under these conditions, the increased permeability of the membrane to ions leads to higher rates of electron flux, and thus respiration (VO_2), as membrane potential decreases. However, if

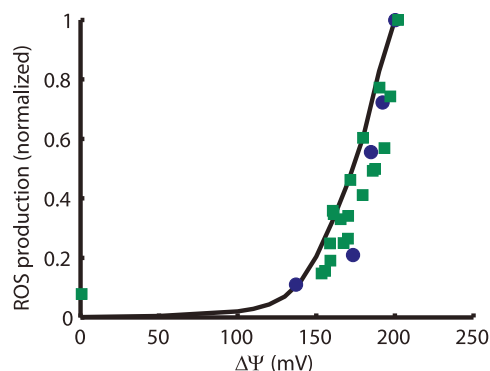


FIGURE 4 $\Delta\Psi_m$ dependence of complex-III-derived ROS. Model ROS production from complex III (black) is minimal for $\Delta\Psi_m < 150$ mV, then increases exponentially as $\Delta\Psi_m$ increases to >150 mV. Experimental data from Korshusnov et al. (18) (circles) and from Rottenberg et al. (17) (squares) follow a similar trend. The normalized $\Delta\Psi_m$ reported by Korshusnov et al. (18) were scaled to a maximum of 200 mV for comparison.

membrane potential is modulated by inhibition of the ETC either by using malonate (18,47) or by modulating the concentration of cytochrome *c* as an electron acceptor (48), then VO_2 will decrease with membrane potential, thus paralleling the trend in ROS production (not shown). Decreasing $\Delta\Psi_m$ using an uncoupler would also be expected to oxidize the NAD^+/NADH couple, which is evident in experiments (9,49) but is not represented in the model due to the exclusion of the TCA cycle. Given that uncoupling oxidizes the cellular redox environment, the decrease in ROS shown in Fig. 4 is consistent with the R-ORB hypothesis over the range of intermediate to highly reduced redox environments.

Model simulations in which matrix pH is increased (Fig. S2 in the Supporting Material) show that ROS production from complexes I and III during FET increases with matrix alkalinization, in agreement with the results of Selivanov et al. (50). From a control matrix pH of 7.3, the model results show a 3.7-fold increase in net ROS production with 0.6 pH units of alkalinization and a 30% decrease with 0.2 units of acidification. As Fig. 5 shows, RET-derived ROS production from complex I also exhibits matrix pH dependence. Under state 4 conditions with high succinate, matrix alkalinization causes an increase in complex I ROS production, qualitatively similar to data shown in Lambert and Brand (22) (reproduced in Fig. 5 A), though the experi-

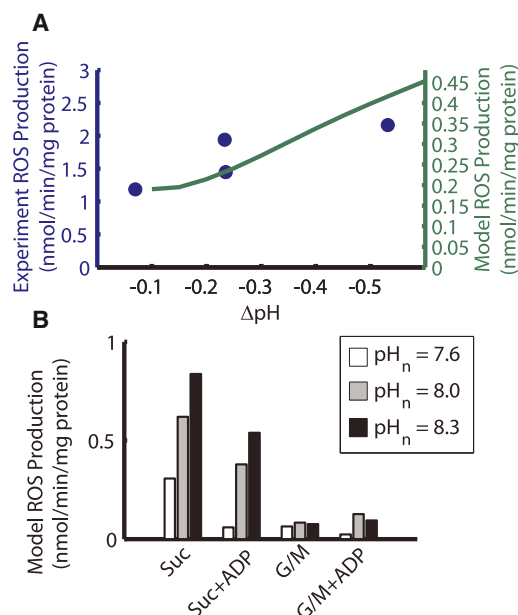


FIGURE 5 Effect of matrix alkalinization on complex I ROS production. (A) Comparison of experimental complex I H_2O_2 production data (circles) from Lambert and Brand (22) and model ROS production for the same ΔpH (line). Model cytosolic pH was held constant at 7.0 and model $\Delta\Psi_m$ was clamped to 200 mV. Experimental data were given with the effect due to $\Delta\Psi_m$ subtracted (22). (B) Model complex I ROS production for an increase in matrix pH from 7.6 (white) to 8.0 (gray) to 8.3 (black). Not only does matrix alkalinization increase complex I ROS for succinate respiration without ADP, it increases ROS for succinate respiration with ADP fivefold at pH 8 and nearly sevenfold at pH 8.3.

mental ROS production rates are around sevenfold greater. The results of Figs. 5 B and S2 show that complex I ROS production under glutamate/malate (G/M) substrates is consistently low, even for high matrix pH. However, a notable prediction of the model is that under state 3 respiration, in the presence of succinate, an increase in matrix pH will elevate complex-I-derived ROS from the minimal levels observed at control pH to the high levels observed in state 4 RET (Fig. 5 B).

The highest levels of ROS production are typically measured in the presence of various ETC inhibitors that promote high levels of reduction at sites favoring direct electron transfer to molecular oxygen. The simulations shown in Fig. 6 predict the effects of these inhibitors on ROS production under different respiratory conditions. Rat heart mitochondria data by St. Pierre et al. (20) show that complex I ROS production is limited to the matrix side, in agreement with the ROS production mechanism chosen for the model (Fig. 6 A). Complex-III-derived ROS production rates in the experiments and the model are low for succinate respiration, except in the presence of antimycin A, which blocks

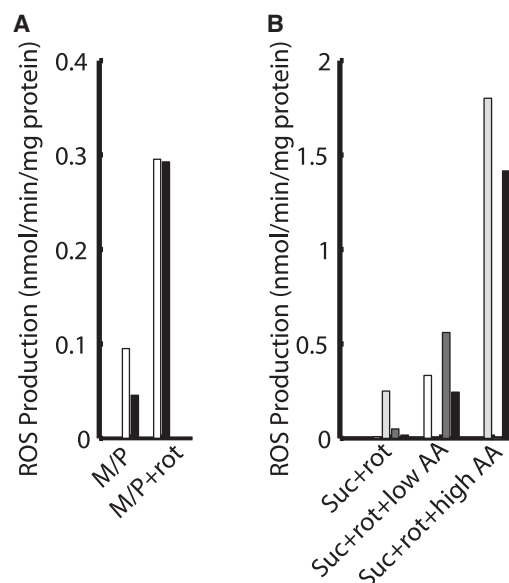


FIGURE 6 Control of ROS production by respiratory blockers. (A) ROS release in the presence of complex I substrates. For mitochondria respiring on malate plus pyruvate (M/P), experimental H_2O_2 recordings (white) from St. Pierre et al. (20) were measured in the absence and presence of rotenone (rot) and compared to model ROS production under similar conditions (black). (B) ROS release from complex III. Experimental results from St. Pierre et al. (20) (white), Turrens et al. (48) (light gray), and Starkov and Fiskum (82) (dark gray) show ROS release for rat heart mitochondria respiring on succinate (Suc). St. Pierre et al., Turrens et al. and simulations results included rotenone to eliminate RET-induced complex I ROS release. Antimycin A (AA) was added to observe ROS production resulting from the disruption of the complex III Q cycle. Similar results are obtained from the model (black). St. Pierre et al. used 0.625 nmol/mg protein AA, Turrens et al. used 2 nmol/mg protein AA, and Starkov and Fiskum used 2 μM AA. Specific model simulation conditions are given in the Supporting Material.

the Q_i site, preventing the b-hemes from being oxidized by matrix-facing Q (Fig. 6 B). This results in an accumulation of cytosol-facing ubisemiquinone and subsequent electron transfer to O_2 and superoxide formation. In a similar way, block of b_L heme reduction by ubisemiquinone at the Q_o site also prevents electrons from crossing the membrane via the b-hemes (not shown). The result is a buildup of cytosol-facing ubisemiquinone and elevation of ROS production. Additional simulations show that inhibition of oxygen binding to complex IV by cyanide decreases ROS production from complex III (not shown) by causing nearly complete reduction of cytochrome *c* and the FeS cluster, preventing the oxidation of QH_2 and formation of semiquinone. These results are in agreement with experimental data demonstrating a cyanide-dependent decrease in ROS production in heart mitochondria (19,51). Each inhibition simulation is the result of drastically reducing or eliminating the reaction in the model analogous to that inhibited by the drug intervention.

Fig. S1 shows the NADH dependence of complex I ROS production in the absence of Q and QH_2 , as in Pryde and Hirst (15). This was predicted by fitting the rate constants of complex I to the data from Aon et al. (9), as shown in Fig. 3. The midpoint potential of this curve matches the two-electron reduction potential of complex I's FMN, given by Sled et al. (53) as -0.38 V at pH 7.5. This prediction supports the validity of the proposed mechanism of ROS production from the FMN (14,15).

Control of ROS levels

Many experiments infer matrix ROS levels from data obtained using indicators of hydrogen peroxide (H_2O_2) emission into the extramitochondrial space (e.g., the Amplex Red method). This technique thus reports only the excess H_2O_2 diffusing across the inner mitochondrial membrane (IMM) at steady state, which reflects the net balance of superoxide production and its removal via superoxide dismutase and glutathione/thioredoxin-dependent H_2O_2 scavenging systems. Thus, the ROS production rates depicted in Fig. 3 provide only one-half of the picture. To more accurately represent mitochondrial ROS dynamics, both ROS production and scavenging must be taken into account. Hence, we combined the ETC ROS production model with a recent minimal model of ROS scavenging (33). Model NADH levels were derived from an earlier mitochondrial model (54) subjected to the same uncoupling protocol. NADPH levels were set to follow the same decreasing function as NADH, but with a minimum of $3 \mu M$ and a maximum of $8 \mu M$, spanning the critical ROS overflow range given in Aon et al. (33). The mitochondrial redox environment was calculated as the sum of redox potentials for NADH, NADPH, and GSH weighted by their respective concentrations, as described in Schafer and Buettner (55). In the simulation, mitochondrial respiratory uncoupling from state 4 is represented by increasing membrane proton leak

conductance, leading to oxidation of the mitochondrial redox environment (Fig. 7). The model's rate of H_2O_2 emission across the IMM decreases from initial levels near $10 \text{ nmol/min/mg protein}$ to a minimum of $\sim 7 \text{ nmol/min/mg protein}$ as the redox environment is oxidized from -1300 to -1150 mV-mM . As uncoupling progresses, the scavenging systems are compromised by further oxidation of the mitochondrial redox environment, and H_2O_2 emission begins to increase with further uncoupling. At -400 mV-mM redox environment, NADH is 2% reduced, $\Delta\Psi_m$ has decreased to 135 mV and the rate of H_2O_2 emission reaches nearly $40 \text{ nmol/min/mg protein}$. This fourfold increase in H_2O_2 emission at -400 mV-mM redox environment compared to the initial H_2O_2 emission rate is in good agreement with the experimental data given in Aon et al. (9).

The model mechanism that produces the U-shaped dependence of ROS levels on redox environment follows closely that proposed in the R-ORB hypothesis. In the model simulations, state 4 conditions are modulated with increasing respiratory uncoupling. Increasing leak causes a decrease in $\Delta\Psi_m$, which drives a subsequent decrease in ROS production, similar to the behavior depicted in Fig. 4. As membrane potential declines from the initial state 4 value, the $NAD^+/NADH$ redox couple becomes more oxidized as more NADH is consumed to pump protons in a futile attempt to counteract the increased membrane leak. ROS scavenging systems remain largely intact for

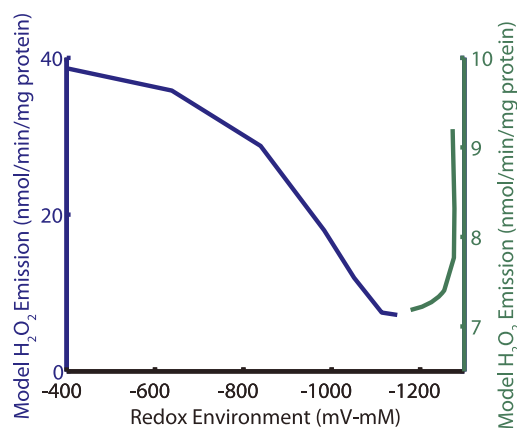


FIGURE 7 Interplay of ROS production and scavenging systems. Simulations were performed with progressive respiratory uncoupling simulated by increasing proton leak conductance in the model. Steady-state H_2O_2 emission rates are plotted against the model's redox environment, incorporating NADH, NADPH, and GSH redox couples as described by Schafer and Buettner (55). The U-shaped dependence of H_2O_2 emission on $\Delta\Psi_m$ arises from the combined effects of decreased ROS production and decreased ROS scavenging capacity on net ROS levels. Model NADH levels were derived from a previously published mitochondria model (54) subjected to the same uncoupling protocol. NADPH levels were set to follow the same decreasing function as NADH, but with a minimum of $3 \mu M$ and a maximum of $8 \mu M$, spanning the critical ROS overflow range given in Aon et al. (33). See the Supporting Material for scavenging equations and parameters.

small oxidative perturbations in the redox environment (not shown). Thus, for the initial increase in uncoupling, comprising the right side of the U-shape in Fig. 7, model ROS production decreases 60%, whereas GSH and the associated scavenging capacity are almost unchanged, leading to the ROS minimum at the so-called redox-optimum. Model $\Delta\Psi_m$ at this redox optimum is 152 mV, similar to the nearly minimal ROS production at 153 mV observed by Korshunov et al. (18) (Fig. 4, *green squares*) Further uncoupling continues to decrease $\Delta\Psi_m$, but ROS production only decreases by another 13% from control levels. The NAD^+/NADH redox couple continues to oxidize, leading to lower NADPH levels and a decreased ability to recharge the oxidized glutathione and thioredoxin scavenger pools. As a result, GSH levels drop by 72%, compromising the H_2O_2 scavenging capacity of the mitochondria. The decrease in GSH levels exceeds the reduction in ROS production, causing an increase in H_2O_2 emission as the redox environment is further oxidized from the optimum.

DISCUSSION

In this work, we have used a mechanistic formulation to develop what is to our knowledge a novel model of the mitochondrial ETC, which we then used to study control of ROS production in cardiac mitochondria. This model is shown to reproduce 1), experimental data measuring the $\Delta\Psi$ dependence of the redox potential of the Q pool and the oxidation states of the b-hemes and cytochromes c_1 and c ; 2), experimental results describing the dependence of the respiratory flux from complex I or complex II substrates on $\Delta\Psi_m$; 3), measurements of ROS production as a function of substrate concentration and $\Delta\Psi_m$. The model predicts the dependence of ROS production on $\Delta\Psi$, matrix pH, NAD^+/NADH redox potential, and inhibition of complexes I and III. With the addition of a minimal ROS scavenging model, the computational work presented here also permits analysis of the relative roles of ROS production and scavenging on mitochondrial H_2O_2 emission and ROS balance, as well as the impact of alterations in the redox environment on oxidative stress.

The detailed description of redox-dependent electron transport processes in the model is the basis for its predictive power. Because the $\Delta\Psi$ dependence of the electron transfer between hemes b_L and b_H is included explicitly, suppression of this flux (e.g., by antimycin A) leads to a cytosolic-side accumulation of ubisemiquinone, producing the $\Delta\Psi$ -dependent ROS production behavior shown in Fig. 4. The complex III inhibitor effects shown in Fig. 6 also arise from the explicit representation of electron transfers of the Q cycle. The antimycin A block increases ROS production by a mechanism similar to the increase produced due to high $\Delta\Psi$. Although the model of complex I does not include explicit electron transfer reactions like the complex III model, each state transition does have a mechanistic basis

(Fig. 1 B). As a result, the complex I model is able to predict the influence of rotenone (Fig. 6 A) and NAD^+/NADH redox potential (Fig. S1) on ROS production.

The explicit inclusion of reactions requiring protons leads to some interesting predictions regarding the pH dependence of ROS production. As shown in Fig. S2, ROS production by complex I and complex III increases under conditions of matrix alkalinization, in agreement with data from Selivanov et al. (50). In the model, the reduction of semiquinone and the binding of two protons are lumped into a single reaction. As matrix-side protons become more scarce, matrix-side ubisemiquinone is converted to ubiquinol more slowly because of the scarcity of protons to form QH_2 . An increasing proportion of the Q pool stagnates on the matrix side without enough protons to become the charge-neutral QH_2 form that is able to diffuse across the membrane, reducing the amount of ubiquinone able to accept electrons from heme b_H . Through these mechanisms, both hemes become more reduced, preventing heme b_L from accepting electrons from the cytosol-side semiquinone. The buildup of this ubisemiquinone species is directly proportional to ROS production from complex III. The model shows that complex I ROS production is also increased under more alkaline matrix conditions. The pH dependence of complex I ROS production is compared to RET data from Lambert and Brand (22) in Fig. 5 A. The model is able to qualitatively reproduce the experimental data. However, a quantitative comparison of model results and experimental data for the pH dependence of complex I shows that the rate of ROS production in the experimental data is around sevenfold higher. ROS production rates vary widely depending on the species used, the age of the animal, and the preparation of the sample. The complex I model was constrained using the data from Aon et al. (9), which show a sixfold-lower rate of ROS production compared with the Lambert and Brand data (22) when mitochondria are respiring on succinate in the absence of ADP. A sixfold increase in the rate of ROS production from complex I (represented by the rate constant between complex I reaction states 4 and 2) can match the magnitude of the Lambert and Brand (22) data without changing the model response to the VO_2 versus $\Delta\Psi$ protocol from Fig. S1 A (not shown), but at the expense of the quantitative agreement shown in Fig. 3. The effects of matrix alkalinization on complex I ROS production were also tested for complex I and complex II substrates in states 3 and 4 (Fig. 5 B). An increase in matrix pH leaves G/M ROS production largely unchanged in state 3 and state 4, but large increases are predicted for succinate respiration. The high levels of complex I ROS production occurring during RET conditions at control pH increase by 87% as pH increases from 7.6 to 8.3. An even larger increase is shown for state 3 succinate respiration, with complex I ROS production increasing almost sevenfold. To our knowledge, this property of state 3 succinate respiration has not been tested experimentally, but this

prediction illustrates the utility of the cycle of experiment-based modeling and model-derived predictions as the basis of further experiments.

The most complete description of mitochondrial ROS levels includes not only ROS production but the native mitochondrial ROS scavenging capabilities. A previously published, minimal model of ROS scavenging (33) was combined with the ROS production model developed here to examine the overall control of mitochondrial ROS levels by both production and scavenging systems. To simulate the oxidation of the mitochondrial environment, respiration was uncoupled from $\Delta\Psi$ with increasing membrane proton leak conductance. In vitro, this has the effect of decreasing NADH levels and subsequently NADPH levels, as the two are coupled via the nicotinamide nucleotide transhydrogenase. Model simulations show that whereas ROS production drops off dramatically as $\Delta\Psi$ decreases (Fig. 4), the depletion of scavenging systems with increased uncoupling results in an increase in H_2O_2 levels as the GSSG/GSH redox couple is oxidized. This is in agreement with the hypothesis put forth by Aon et al. (9) that the interplay between ROS production and ROS scavenging leads to a minimum of mitochondrial ROS levels at intermediate redox potentials. The simulations presented using this model make it possible to explore a range of redox environments both more reduced and more oxidized than that of the ROS minimum.

Validation of the redox-optimized ROS balance hypothesis

By combining the model of ROS production by the ETC, presented here, and the previously published minimal scavenging model (33), we are able to computationally validate the R-ORB hypothesis. In simulations, the mitochondrial redox environment was modulated by increasing respiratory uncoupling, starting from state 4 conditions. The presence of a minimum in H_2O_2 emission at an intermediate redox environment and the 1:4 ratio of H_2O_2 emission for the maximally reduced environment in this protocol compared to the most oxidized environment is in good agreement with the experimental data of Aon et al. (9) H_2O_2 emission reaches a minimum at levels of uncoupling corresponding to a $\Delta\Psi_m$ between 150 and 155 mV, similar to state 3 $\Delta\Psi_m$ values for isolated mitochondria (9) and to the $\Delta\Psi_m$ value shown by Korshunov et al. (18) to strongly suppress H_2O_2 emission. However, this $\Delta\Psi_m$ range is higher than the typical mitochondrial operating conditions in vivo, which are closer to 100–130 mV (43). Therefore, the expectation is that intact cells operate at the optimal conditions for minimizing ROS levels during high rates of oxidative phosphorylation. It is important to note that the model simulates the behavior of isolated mitochondria, which can be pushed to redox extremes and utilize substrate concentrations not normally encountered in cells. The difference in operating con-

ditions between isolated myocytes and isolated mitochondria was a key point addressed by the R-ORB hypothesis (9). Moreover, a whole-cell ROS model would also need to take into account other nonmitochondrial ROS sources (e.g., NADPH and xanthine oxidases) and cytoplasmic ROS scavenging pathways. Although the model $\Delta\Psi_m$ value at the redox optimum is reasonable, the state of the model redox environment at that optimum is far more reduced than that of the experimentally-observed optimum (9). This discrepancy may be due to the approximation of NADH and NADPH levels used in the model. Specifically, the simplifying assumption that NADPH decreases in parallel with NADH may not be an accurate approximation. However, given that the work presented here represents ROS production and scavenging networks in the absence of key NADH and NADPH production mechanisms such as the TCA cycle or the nicotinamide nucleotide transhydrogenase, the qualitative agreement between the model responses to redox environment changes and the experimental data are still highly informative with regard to how ROS production is modulated and how ROS production contributes to matrix ROS levels. The ROS production results shown in this protocol incorporate changes in both $\Delta\Psi$ and NADH redox state that occur with uncoupling. To continue the cycle of experiments and modeling, the ROS production results shown here could be supported by experiments using antioxidant-depleted mitochondria and valinomycin or nigericin to investigate changes in ROS production with $\Delta\Psi$ and ΔpH and to validate the role of $\Delta\Psi$ versus NADH redox state in controlling ROS production, as suggested by the model. Once the model has been validated by additional experiments, it can be used to investigate the impact of $\Delta\Psi$ and NADH redox state on scavenging and ROS production, respectively.

This model, by extension of the R-ORB hypothesis, shows the importance of the interplay between ROS production and scavenging. The uncoupling protocols reproduced with the model (Fig. 7) show the effects of oxidizing the mitochondrial redox environment. ROS production decreases as NADH is oxidized and $\Delta\Psi$ decreases, but net ROS levels begin to increase as GSH becomes oxidized. An interesting extension of this hypothesis would be to investigate changes in ROS production and scavenging that result from a more reduced redox environment, as might occur during inhibition with cyanide. Although the model presented here does not include a dynamically varying redox environment, preliminary studies show that, with $\Delta\Psi$ and NADH redox potential clamped, application of submaximal cyanide block increases ROS production (not shown), in agreement with data from the literature, which also show reduced complex IV function and incomplete block of respiration (48,56,57). In contrast to the oxidation associated with uncoupling, the inhibition of respiration by cyanide will serve to reduce the electron carriers of the ETC. In the uncoupling protocol, oxidation of the redox

environment depletes the mitochondrial scavenging capacity and leads to ROS overflow. Anoxia protocols would explore reduction of the redox environment, in which case the R-ORB hypothesis suggests that ROS production will play a larger role than scavenging in ROS overflow.

Mechanism of ROS production from complex I

Complex I ROS production was modeled as originating from a redox center in the hydrophilic arm of the protein. Without the explicit inclusion of complex I redox centers, this is a generalized implementation of the hypothesis, proposed by Turrens and Boveris (19) and further investigated in recent years by the Hirst group (14,15), that complex I ROS is produced from the FMN site. Under this assumption of complex I ROS production, the model can reproduce experimental data describing ROS production using complex I and complex II substrates under state 3 and state 4. In the absence of ubiquinone or ubiquinol, the model predicts a dependence of ROS production on NADH redox potential, closely matching experimental data from Pryde and Hirst (15) (see Fig. S1). In the work by those authors, it is noted that the midpoint potential of this NADH dependence closely matches the two-electron reduction potential of the complex I FMN. This serves as validation of the mechanism chosen in the model to represent complex I ROS production. Other groups (22,28,58) have proposed complex I's quinone-binding site to be the primary source of ROS production in the absence of rotenone. The model described here was modified to test this Q-linked ROS production hypothesis by removing the superoxide-generating reaction connecting states 4 and 2 of the complex I model (see Fig. 1 B) and replacing it with a reaction connecting state 7 (reduced enzyme bound to Q) to state 2, such that the ROS production cycle of the revised model connects states 2, 3, 4, 7, and 2 again (see the Supporting Material for details). When the modified model was refitted to the ROS production data from Aon et al. (9), as for the original model, the ability of the Q-linked ROS production model to reproduce the data were very similar (not shown). None of the parameters for the Q-site ROS production model varied by more than 10 times the original model parameters, with the largest differences occurring in the rates for NADH and NAD^+ binding. Given that the modeling results for the two different hypotheses were identical, this modeling experiment was unable to distinguish between the two mechanisms.

Mechanism of ROS production from complex III

The model presented here assumes that the mechanism of ROS production from complex III involves the reduction of oxygen by the cytosol-side semiquinone radical and is associated with a highly reduced ubiquinone pool and high proton motive force. Since the Demin et al. models

(34,35) were developed, providing the basis for the model presented here, more recent experimental data have led to alternative proposals for complex-III-derived ROS production. Dröse and Brandt (59) found that in submitochondrial particles, ROS production is maximized when the Q pool is 25% oxidized, in contrast to the mechanism in the model presented here, which requires a highly reduced Q pool. These data led Dröse and Brandt to propose a mechanism of ROS production derived from RET from the reduced heme b_L to oxidized ubiquinol to form a transient semiquinone, which reduces oxygen to form superoxide. Both this mechanism and the one used in the model depend on highly reduced heme b_L derived from high $\Delta\Psi$. The difference arises from whether the electron that forms superoxide is released from the forward reaction from the QH_2 bifurcation with FeS or in the reverse reaction from heme b_L . Another recently proposed mechanism comes from Borek et al. (60), who used mutant, bacterial complex III proteins to study the impact of the position of the FeS head domain on ROS production. Their results show that ROS production occurs when the FeS head is in the distal position, when the diffusion of oxygen to a position near enough for electron transfer from the Q_o site is not sterically hindered. Those authors also suggest that the highest ROS production rates may occur when a semiquinone is formed at the Q_o site via RET from heme b_L . Both of these mechanisms, in contrast to the mechanism used in this model, emphasize ROS production without a significant accumulation of semiquinone at the Q_o site. All of the mechanisms described rely on reduced heme b_L to produce ROS. However, the differences in the distribution of ubiquinone forms and origin of the electron used to generate the semiquinone at Q_i may lead to different $\Delta\Psi$ dependence of the electron carrier redox states. The ROS production mechanism of the model is subject to further refinements as consensus is achieved in the bioenergetics community with regard to the mechanism of ROS production in complex III.

Critique of the model

The model presented here was constructed with the goal of incorporating a high level of mechanistic detail while limiting the number of equations necessary. This will be an asset in future work, when the model of respiration is to be incorporated into a full description of mitochondrial dynamics and eventually further nested into a cardiac myocyte model. Such a multiscale model will require a conservative number of states and a limited range of timescales for computational efficiency.

The model presented here only accounts for superoxide release from the Q_o site. Given that the difference in redox potential between the ubiquinone/semiquinone and semiquinone/ubiquinol couples is significantly smaller at the Q_i site than at the Q_o site (61), the semiquinone at Q_i is considered thermodynamically stable (30,62). As such, we

have not accounted for superoxide release from that radical. However, it is worth noting that the concentration of semiquinone at Q_i does increase with membrane potential as well as matrix alkalization (not shown), following a trend similar to that observed for the superoxide-producing semiquinone at Q_o .

The complex IV model included here is a relatively simplistic description of an enzyme with a wide variety of properties, from allosteric ATP inhibition to variable K_m for O_2 to intrinsic uncoupling at high $\Delta\Psi_m$. A recent model by Krab et al. (63) incorporates more mechanistic detail to model the changing K_m . This could be a potentially valuable addition to the model described here, facilitating investigation of ROS production in conditions of varying oxygen content, such as in a spatial model or under an ischemia-reperfusion protocol.

As described above, mitochondrial and cellular ROS levels are determined by both ROS production and ROS scavenging. Only the simulations of Fig. 7 include the contribution of mitochondrial ROS scavenging. As such, our model underreports the amount of ROS produced in most protocols. The most comprehensive estimates would require a full mitochondrial model that includes NADH and NADPH production, which is outside the scope of this work.

The mitochondrial ROS production and scavenging pathways represent a highly coupled and nonlinear system. The goal of this modeling effort was to reproduce key experimental protocols with a minimal model of mitochondrial regulation and control. Although the model's ability to reproduce some of the experimental data falls short, as shown by the disparity in cytochrome *c* oxidation in Fig. 2 C, these disparities should not be seen as failures, but instead as clues to critical control mechanisms that are not represented in this version of the model.

Comparison to existing models

Several models reported in the literature describe the ETC. Some are detailed representations of electron transfer reactions (64–67), with the advantage of mechanistic accuracy at the cost of computational intensity. Others use a simpler thermodynamic approach (37,68) that requires fewer equations and state variables, but may capture less detail in the model results. For a full comparison of the work presented here with previously published modeling efforts, please refer to the Supporting Material.

CONCLUSION

A mechanistic approach to modeling the mitochondrial electron transport chain led to the construction of a model that displays a strong qualitative agreement with experimental data describing the control of respiration of ROS production and good quantitative agreement with many of

those protocols. An additional asset of the model afforded by its mechanistic nature is the ability to reproduce experimental data describing the influence of inhibitors on ROS production. Furthermore, the model offers predictions regarding complex I ROS generation mechanisms and the interplay between ROS production and scavenging that together set mitochondrial and cellular ROS levels. This model has not only strong explanatory value but the ability to predict properties of the ETC in the protocols simulated in this study and potentially in further studies.

SUPPORTING MATERIAL

Four figures, 17 tables, references (69–80) and Supporting Text, are available at [http://www.biophysj.org/biophysj/supplemental/S0006-3495\(13\)00788-1](http://www.biophysj.org/biophysj/supplemental/S0006-3495(13)00788-1).

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**A Computational Model of
Reactive Oxygen Species and Redox Balance in Cardiac Mitochondria**

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Methods

Minimal Mitochondrial Model

In order to represent the state-dependent changes in inner membrane potential that occur upon addition of substrate and ADP to substrate-depleted mitochondria, a model of proton leak flux and ATPase were incorporated, based on a previously published model (1) with parameters as given in the supplement. A further addition of a minimal model of ROS scavenging in the mitochondrial matrix, as published in Aon et al. (2) was added to investigate the interplay of ROS production and scavenging.

Computational Methods

The combined ETC model introduced here consists of 14 nonlinear ordinary differential equations describing the oxidation state of 6 Q species on both sides of the membrane, the coupled b_L - b_H redox state, the complex III FeS cluster, cytochrome c_1 and cytochrome c. The equations were implemented in MATLAB 7.1 (The MathWorks, Natick, MA). Model parameters were optimized using the `fmincon` function from MATLAB. For the protocols described, the model was run until steady-state, when the magnitude of each time derivative was $\leq 10^{-8}$.

Results

Control of Respiration

The activity of the ETC is driven by membrane potential, matrix and cytosolic pH and substrate concentration. Fig. S3 shows the $\Delta\Psi_m$ -dependence of model respiration on NADH (Fig. 3A) and succinate (Fig. 3B) compared to experimental data. This dependence follows the characteristic Z-shape (3) with high, saturating respiratory flux for low $\Delta\Psi_m$ decreasing through a linear region for $\Delta\Psi_m$ between state 3 and state 4 values to minimal respiration for $\Delta\Psi_m$ in the high or higher than physiological range. For decreasing ΔpH , expressed as the cytosolic pH less the matrix pH, the V_{O_2} vs. $\Delta\Psi_m$ curve shifts leftward (not shown), as in the Magnus and Keizer ETC model(4). Substrate dependence for complex I substrates is validated by comparison with the results of isolated mitochondria undergoing a state 0-4-3 transition protocol. In order to simulate this protocol, the ETC model was expanded into a minimal mitochondrial model with the ability to track changes in $\Delta\Psi_m$ by adding a leak current and a model of the mitochondrial F_0 - F_1 ATPase, as previously published (1). The experimental protocol of Wei et al. (1) (their Fig. 4) was simulated in Fig. S4 in the Supporting Material by adjusting NADH levels according to values measured in that experiment and by varying ADP levels accordingly. Model $\Delta\Psi_m$ for all three states is in good agreement with the experimental data from guinea pig cardiac mitochondria, demonstrating appropriate control of model respiration by $\Delta\Psi_m$ and NADH levels.

Supplemental Figures

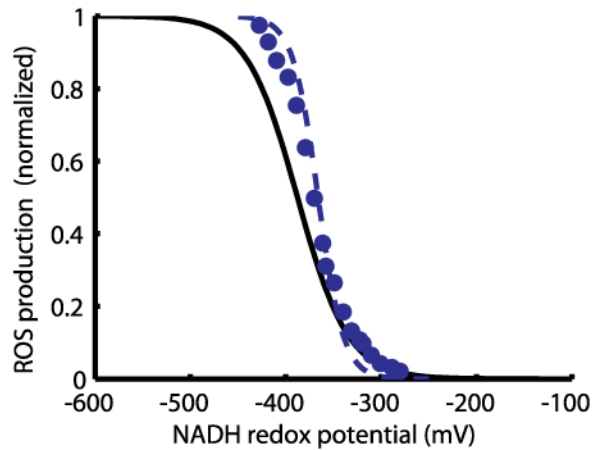


FIGURE S1 Redox potential dependence of complex I-derived ROS.

ROS production from the isolated complex I model was simulated for a range of NADH redox potentials (*solid line*). These results were compared to experimental data (*circles*) from Pryde and Hirst (8) and their fit (*dashed line*)

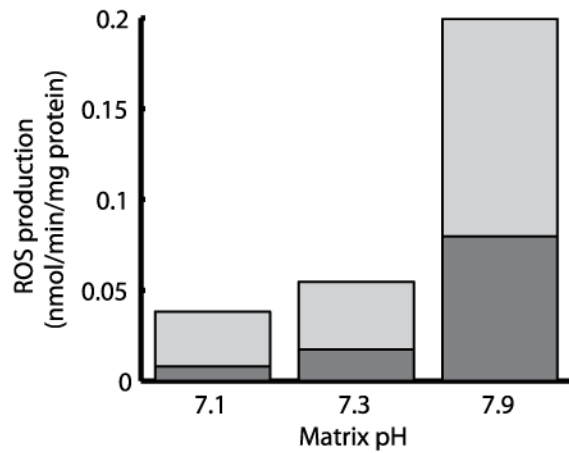


FIGURE S2 Effect of matrix pH on model ROS production under state 4 FET conditions.

ROS production was simulated in glutamate/malate (G/M) in the absence of ADP. Bottom bars (*dark gray*) depict ROS production originating from complex I, top bars (*light gray*) depict matrix-side ROS from complex III. pH 7.3 is representative of the control pH used in other simulations.

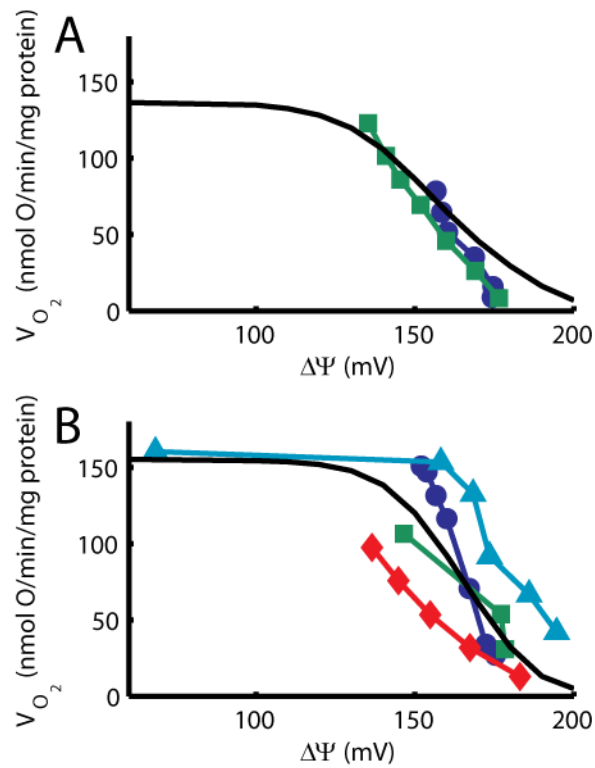


FIGURE S3 $\Delta\Psi_m$ -dependence of respiratory rate on complex I and complex II substrates.

(A) Model results (*black*) follow closely the experimental measurements from Marcinkeviciute (5) (*circles*) and Lionetti et al. (6) (*squares*), both recorded from isolated mitochondria respiring on glutamate and malate. (B) Model simulations (*black*) of mitochondrial respiration on succinate compared with experimental data reproduced from Marcinkeviciute et al. (5) (*circles*) and Lionetti et al. (6) (*squares*), Hafner et al. (7) (*diamonds*), and Murphy and Brand (*triangles*).

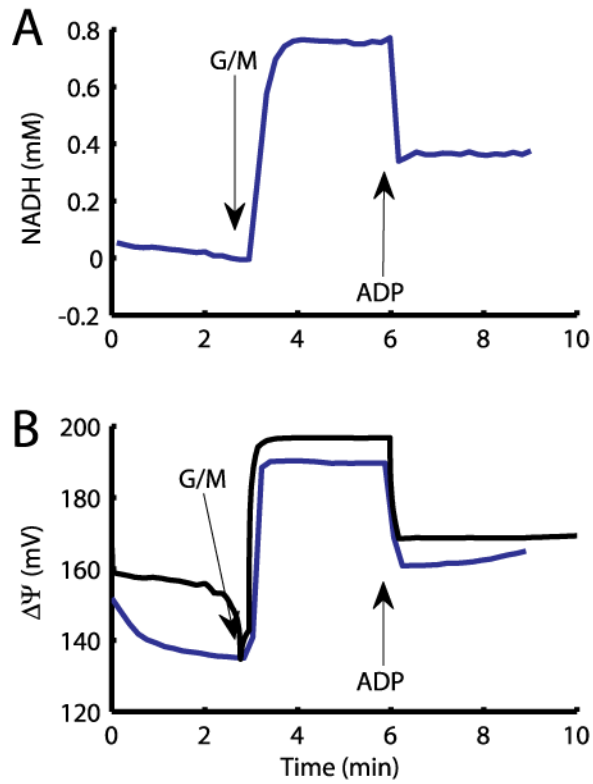


FIGURE S4 Mitochondrial respiratory state transitions.

(A) Experimental NADH recordings from Wei et al. (1) in isolated guinea pig heart mitochondria. These data were used as input to the model to simulate a protocol of metabolic transitions from state 1 to state 4 (addition of substrate glutamate/malate) to state 3 (addition of ADP). (B) Model $\Delta\Psi_m$ (black) closely follows the corresponding experimental measurements (1) of membrane potential (blue).

Discussion

Comparison to Existing Models

Computational models must compromise between highly detailed, mechanistic descriptions that are computationally intensive and more phenomenological representations that are more tractable. The model presented here was built while trying to minimize the number of state variables. However, several more detailed models of the ETC exist. The extension of the 1998 Demin model upon which the complex IV model included here was based describes the complete redox state of complex III (b_L , b_H , FeS and cytochrome c_1) and adds the restriction that the semiquinone at the Q_O site does not freely diffuse, but remains localized to the Q_O binding pocket. While this adds more mechanistic detail, it also increases the minimal number of states required to describe the complex III redox state from 3 to 12. This more realistic description of the semiquinone may better reflect the quantitative details of ROS production, but the simplified model used here successfully reproduces the experimentally demonstrated effect of $\Delta\Psi_m$ on complex III ROS production with less computational cost. A series of models has been published by Selivanov et al. (8-10) featuring a highly detailed description of electron transfer and utilizing an innovative algorithm to automate the construction of the ODEs describing the model's redox states. The detailed description of protein redox states in the group's modeling including complex I (9, 10) offers unique insight into the feasibility of the semiquinone-derived complex I ROS production theory. However the 10 states describing complex I and 400 states describing complex III likely preclude the incorporation of this model in its present form into a full mitochondrial model. Orii and Miki (11) published another detailed complex III model including 56 redox states of the enzyme. This model captures to a high degree the complex kinetics the authors observed in their spectrophotometric studies. The combined results of their experimental and modeling studies offer interesting insights into the electron bifurcation between the FeS and b_L redox centers. However, their kinetic studies span durations of three seconds or less. Simulations of metabolic state transitions, during which NADH levels reach steady-state over the course of minutes, might pose a prohibitive computational challenge for this model. More generalized models of the ETC have also been published by Magnus and Keizer(4) and Jin and Bethke(12). The former is the basis for the complex I model included here. The equations describing the various states of the lumped respiratory chain in the Magnus and Keizer ETC model(4) are algebraic, making it an excellent choice for incorporation into integrated models, such as the 2003 mitochondrial energetics model of Cortassa et al. (13) However, because the Magnus and Keizer model (4) is built on a description similar to the ETC supercomplex (14), the model does not include descriptions of the electron carrier intermediates ubiquinone/ubiquinol and cytochrome c. Thus the Cortassa et al. model (13) does not allow determination of ROS production. Jin and Bethke(12) published a more recent ETC model built on a similar non-equilibrium thermodynamics foundation. Theirs is more generalized, allowing for the simulation of respiration on any combination of complex I and complex II substrates, as described by the redox potential of the $NAD^+/NADH$ and succinate/fumarate couples. However, it would be equally difficult to augment this model with an expression for mechanistic ROS production. The mitochondrial and cellular models of ROS-induced ROS release (RIRR) by Cortassa et al. (15) and Zhou et

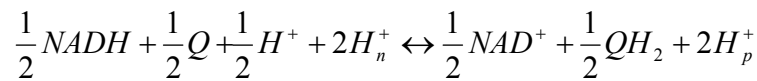
al. (16), respectively, contain an ad hoc “shunt” description of ROS release as a constant proportion of respiratory flux. As is shown in Fig. 7, there are conditions under which ROS production increases with increases in $\Delta\Psi_m$ and thus follows the opposite trend of respiratory flux. As such, the shunt model cannot account for the control of ROS production in all scenarios. The incorporation of a more mechanistic ETC model, such as the one presented here, will lead to a more complete description of the ROS production component of the RIRR phenomenon.

Modeling Methods

Model code is available online at <http://icm.jhu.edu/models>.

Section 1. NADH-Ubiquinone Oxidoreductase (Complex I)

A model of complex I was constructed based on the non-equilibrium thermodynamic description from Magnus and Keizer’s mitochondrial model(4). To separate the complex I contribution from their lumped ETC model, a state was added in order to incorporate the reduction of Q to QH_2 as the product of complex I. Rate constants were modified from Magnus and Keizer(4) to (roughly) reflect the overall equilibrium constant for the complex I reaction as reported by Chen et al(17). Proton stoichiometry coefficients were reduced to account for the number of protons pumped by complex I only. Proton slip and reaction slip reactions were removed such that oxidation of NADH is completely coupled to proton pumping in the current model. To reduce the state space of the model, Magnus and Keizer represented only the transfer of one electron at a time from NADH. A similar approach was taken here such that the overall reaction is



The rate of production of ubiquinol is then twice the flux through the complex I model. To model ROS production from complex I, a reversible reaction between states 4 and 2 was added to represent the transfer of an electron from a reduced redox center in the matrix localized arm of the protein to oxygen according to the widely-accepted model of complex I-derived ROS (18-20). Fig. 1 B in the main text gives an overview of the reactions represented in the complex I model. The MATLAB tool KA Pattern (21) was applied to the complex I model in order to derive an algebraic expression for the steady-state electron transfer flux through the complex. Complex I rate constants were optimized using respiratory flux and ROS production rates from Aon et al. (22) as well as $NAD^+/NADH$ and Q/QH_2 redox data from Muraoka and Slater (23).

Rate equations used in complex I model

$$a_{12} = a_{12}^* \cdot H_m^2$$

$$a_{65} = a_{65}^* \cdot H_i^2$$

$$a_{61} = a_{61}^* \cdot \exp(-2 \cdot 0.5 \cdot (\Delta\Psi - \Delta\Psi_B) \cdot F / (RT))$$

$$a_{16} = a_{16}^* \cdot \exp(2 \cdot 0.5 \cdot (\Delta\Psi - \Delta\Psi_B) \cdot F / (RT))$$

$$a_{23} = a_{23}^* \text{NADH}^{1/2}$$

$$a_{43} = a_{43}^* (\text{NAD}^+)^{1/2}$$

$$a_{47} = a_{47}^* Q^{1/2} \cdot H_m^{1/2} \cdot (1 - \text{rotenoneBlock})$$

where the other half proton is retained from $1/2\text{NADH}$

$$a_{57} = a_{57}^* \cdot (\text{QH}_2)^{1/2} \cdot (1 - \text{rotenoneBlock})$$

$$\text{KeqROS} = \exp(2 \cdot (E_{\text{O}_2^-} - E_{\text{FMN}}) \cdot F / (RT))$$

$$a_{42} = a_{42}^* \cdot \text{O}_2$$

$$a_{24} = a_{42} \cdot \text{KeqROS} \cdot \text{O}_2^-$$

$$D = E_1 + E_2 + E_3 + E_4 + E_5 + E_6 + E_7$$

Numerator terms E1 through E7 have 40 terms, each the product of 6 rate constants. For details, see model code available online at <http://www.icm.jhu.edu/models>.

$$F_1 = E_1 / D$$

$$F_2 = E_2 / D$$

$$F_3 = E_3 / D$$

$$F_4 = E_4 / D$$

$$F_5 = E_5 / D$$

$$F_6 = E_6 / D$$

$$F_7 = E_7 / D$$

Jres is the steady-state flux of a single electron through complex I in electrons/min

$$\text{Jres} = 60 \cdot \rho_{\text{C1}} \cdot (F_4 \cdot a_{47} - F_7 \cdot a_{74})$$

v_{c1} is the rate of reduction of Q to QH_2 by complex I, which requires two electrons

$$v_{\text{c1}} = \text{Jres} / 2$$

VROS_{c1} is the rate of reduction of O_2 to O_2^- by complex I in $\text{mM O}_2^-/\text{min}$

$$\text{VROS}_{\text{c1}} = 60 \cdot \rho_{\text{C1}} \cdot (F_4 \cdot a_{42} - F_2 \cdot a_{24})$$

TABLE S1. Parameters used in complex I model

Parameter	Description	Value	Reference
ρ_{C1}	concentration of complex I	8.8490mM	fit using data from (22)
$\Delta\Psi_B$	phase boundary potential	50mV	(4)
a_{12}^*		$6.3396 \cdot 10^{11} \text{ mM}^{-2} \text{ s}^{-1}$	fit using data from (22)
a_{21}		5 s^{-1}	(4)

a ₅₆		100 s ⁻¹	(4)
a ₆₅ *		2.5119·10 ¹³ mM ⁻² s ⁻¹	fit using data from (22)
a ₆₁ *		10 ⁷ s ⁻¹	fit using data from (22)
a ₁₆ *		130 s ⁻¹	(4)
a ₂₃ *		3.8867·10 ³ mM ^{-1/2} s ⁻¹	fit using data from (22)
a ₃₂		9.1295·10 ⁶ s ⁻¹	fit using data from (22)
a ₃₄		639.1364 s ⁻¹	fit using data from (22)
a ₄₃ *		3.2882 mM ^{-1/2} s ⁻¹	fit using data from (22)
a ₄₇ *		1.5962·10 ⁷ mM ⁻¹ s ⁻¹	fit using data from (22)
a ₇₄		65.2227 s ⁻¹	fit using data from (22)
a ₇₅		2.4615·10 ⁴ s ⁻¹	fit using data from (22)
a ₅₇ *		1.1667·10 ³ mM ^{-1/2} s ⁻¹	fit using data from (22)
E _{FMN}	Midpoint potential of flavin mononucleotide	-0.375mV	(24)
E _{O2.-}	Midpoint potential of superoxide	-0.15mV	(25)
a ₄₂ *		6.0318 mM ⁻¹ s ⁻¹	fit using data from (22)

Section 2. Succinate Dehydrogenase (Complex II) and Ubiquinone-Cytochrome c Oxidoreductase (Complex III)

A simplified model of the complex II enzyme was adapted from Demin et al (25). The complex II model represents saturating succinate concentrations and depends only on the ratio of ubiquinol to ubiquinone on the matrix face of the inner membrane. Conditions of varying succinate concentration were simulated by modulating the maximal reaction rate.

The complex III model used here is derived from the representation in Demin et al (25). Briefly, the ratio of forward and backward rate constants for each reaction, as represented in Fig. 1 C of the main text, was defined by experimentally-measured redox potentials as described in Demin et al. (25). Parameters were further constrained by fitting the oxidation state of the redox centers in complex III to experimental data from Kim et al. (26) and Brown and Brand (27).

System of differential equations describing complex III

$$\begin{aligned}
 dQ_n/dt &= v_5 - v_{7_bLox} - v_{7_bLred} - v_1 \\
 dQ_n^-/dt &= v_{7_bLox} + v_{7_bLred} - v_{8_bLox} - v_{8_bLred} \\
 d(QH_2)_n/dt &= v_{8_bLox} + v_{8_bLred} + v_1 - v_2 \\
 d(QH_2)_p/dt &= v_2 - v_3 \\
 dQ_p^-/dt &= v_3 - v_{10} - v_{10b} - v_{4_bHox} - v_{4_bHred} \\
 dQ_p/dt &= v_{10} + v_{10b} + v_{4_bHox} + v_{4_bHred} - v_5 \\
 db1/dt &= v_{7_bLox} + v_{8_bLox} - v_{4_bHox} \\
 db2/dt &= v_{4_bHox} + v_{7_bLred} + v_{8_bLred} - v_6 \\
 db3/dt &= -v_{4_bHred} + v_6 - v_{7_bLox} - v_{8_bLox} \\
 db4/dt &= v_{4_bHred} - v_{7_bLred} - v_{8_bLred} \\
 dFeS_{ox}/dt &= v_9 - v_3 \\
 dcyt\ c1_{ox}/dt &= v_{33} - v_9 \\
 dcyt\ c_{ox}/dt &= v_e - v_{33} \\
 d\Delta\Psi/dt &= (V_{hRes} - V_{hu} - V_{hleak})/C_{mito}
 \end{aligned}$$

Where b1 represents the b-heme state where b_L and b_H are both oxidized; b2 is b_L reduced, b_H oxidized; b3 is b_L oxidized, b_H reduced; b4 is both reduced.

Rate equations for complexes II and III

Reaction 1 -- reduction of Q to QH₂

$$\begin{aligned}
 v_1 &= v_{c1} + v_{c2} \\
 &(\text{see } v_{c1} \text{ above})
 \end{aligned}$$

$$v_{c2} = \frac{V_{SDH} \cdot Q_n / (Q_n + (QH_2)_n)}{K_m + Q_n / (Q_n + (QH_2)_n)}$$

Reaction 2 -- diffusion of QH₂ across the membrane

$$v_2 = k_d \cdot ((QH_2)_n - (QH_2)_p)$$

Reaction 3 -- QH₂ to FeS e⁻ transfer

(Note that this reaction describes the oxidation of QH₂ to QH⁺ followed by non-redox dependent dissociation of QH⁺ into Q and H⁺ such that $\Delta E/\Delta pH = -2.3RT/F$ -- see Schafer et al. (28) for details.)

$$\begin{aligned}
 k_3 &= k_{03} \cdot K_{eq3} \cdot \exp(2.3 \cdot (7 - pH_p)) \\
 k_{-3} &= k_{03} \\
 v_3 &= k_3 \cdot (QH_2)_p \cdot FeS_{ox} - k_{-3} \cdot Q_p^- \cdot FeS_{red} \cdot (1 - \text{myxothiazolBlock})
 \end{aligned}$$

Reaction 4 -- Q_p⁻ to b_H e⁻ transfer

$$\begin{aligned}
 k_{4_bHox} &= k_{04} \cdot K_{eq_4_bHox} \cdot \exp(-\alpha \cdot \delta_1 \cdot F / (RT) \cdot \Delta\Psi) \\
 k_{4_bHred} &= k_{04} \cdot K_{eq_4_bHred} \cdot \exp(-\alpha \cdot \delta_1 \cdot F / (RT) \cdot \Delta\Psi) \\
 k_{-4_bHox} &= k_{04} \cdot \exp(\alpha \cdot (1 - \delta_1) \cdot F / (RT) \cdot \Delta\Psi) \\
 k_{-4_bHred} &= k_{04} \cdot \exp(\alpha \cdot (1 - \delta_1) \cdot F / (RT) \cdot \Delta\Psi) \\
 v_{4_bHox} &= k_{4_bHox} \cdot Q_p^- \cdot b1 - k_{-4_bHox} \cdot Q_p \cdot b2; \\
 v_{4_bHred} &= k_{4_bHred} \cdot Q_p^- \cdot b3 - k_{-4_bHred} \cdot Q_p \cdot b4;
 \end{aligned}$$

Reaction 5 – diffusion of Q across the membrane

$$v_5 = k_d \cdot (Q_p - Q_n)$$

Reaction 6 – b_H to b_L e^- transfer

$$k_6 = k_{06} \cdot K_{eq_6} \cdot \exp(-\beta \cdot \delta_2 \cdot F / (RT) \cdot \Delta\Psi)$$

$$k_{-6} = k_{06} \cdot \exp(\beta \cdot (1 - \delta_2) \cdot F / (RT) \cdot \Delta\Psi)$$

$$v_6 = k_6 \cdot b_2 - k_{-6} \cdot b_3$$

Reaction 7 – b_L to Q_n e^- transfer

$$k_{7_bLox} = k_{07_bLox} \cdot K_{eq_7_bLox} \cdot \exp(-\gamma \cdot \delta_3 \cdot F / (RT) \cdot \Delta\Psi)$$

$$k_{7_bLred} = k_{07_bLred} \cdot K_{eq_7_bLred} \cdot \exp(-\gamma \cdot \delta_3 \cdot F / (RT) \cdot \Delta\Psi)$$

$$k_{-7_bLox} = k_{07_bLox} \cdot \exp(\gamma \cdot (1 - \delta_3) \cdot F / (RT) \cdot \Delta\Psi)$$

$$k_{-7_bLred} = k_{07_bLred} \cdot \exp(\gamma \cdot (1 - \delta_3) \cdot F / (RT) \cdot \Delta\Psi)$$

$$v_{7_bLox} = k_{7_bLox} \cdot Q_n \cdot b_3 - k_{-7_bLox} \cdot Q_n^- \cdot b_1 \cdot (1 - \text{antimycinBlock})$$

$$v_{7_bLred} = k_{7_bLred} \cdot Q_n \cdot b_4 - k_{-7_bLred} \cdot Q_n^- \cdot b_2 \cdot (1 - \text{antimycinBlock})$$

Reaction 8 -- b_L to Q_n^- e^- and proton transfer

(Note that this reaction describes the reduction of Q^- to QH_2 such that $\Delta E / \Delta pH = -2 \cdot 2.3RT / F$.)

$$k_{8_bLox} = k_{08_bLox} \cdot K_{eq_8_bLox} \cdot \exp(-\gamma \cdot \delta_3 \cdot F / (RT) \cdot \Delta\Psi) \cdot \exp(2.3 \cdot 2 \cdot (7 - pH_n))$$

$$k_{8_bLred} = k_{08_bLred} \cdot K_{eq_8_bLred} \cdot \exp(-\gamma \cdot \delta_3 \cdot F / (RT) \cdot \Delta\Psi) \cdot \exp(2.3 \cdot 2 \cdot (7 - pH_n))$$

$$k_{-8_bLox} = k_{08_bLox} \cdot \exp(\gamma \cdot (1 - \delta_3) \cdot F / (RT) \cdot \Delta\Psi)$$

$$k_{-8_bLred} = k_{08_bLred} \cdot \exp(\gamma \cdot (1 - \delta_3) \cdot F / (RT) \cdot \Delta\Psi)$$

$$v_{8_bLox} = k_{8_bLox} \cdot Q_n^- \cdot b_3 - k_{-8_bLox} \cdot (QH_2)_n \cdot b_1 \cdot (1 - \text{antimycinBlock})$$

$$v_{8_bLred} = k_{8_bLred} \cdot Q_n^- \cdot b_4 - k_{-8_bLred} \cdot (QH_2)_n \cdot b_2 \cdot (1 - \text{antimycinBlock})$$

Reaction 9 – FeS to cyt c_1 e^- transfer

$$k_9 = k_{09} \cdot K_{eq9}$$

$$k_{-9} = k_{09}$$

$$v_9 = k_9 \cdot FeS_{red} \cdot cyt\ c1_{ox} - k_{-9} \cdot FeS_{ox} \cdot cyt\ c1_{red}$$

Reaction 10 – superoxide production from Q_p^-

$$k_{10} = k_{010} \cdot K_{eq10}$$

$$k_{-10} = k_{010}$$

$$v_{10} = k_{10} \cdot Q_p^- \cdot O_2 - k_{-10} \cdot Q_p \cdot O_2^-$$

$$v_{10b} = k_{10} \cdot Q_p^- \cdot O_2 - k_{-10} \cdot Q_p \cdot O_2^-$$

Reaction 33 – cyt c_1 to cyt c e^- transfer

$$v_{33} = k_{33} \cdot cyt\ c1_{red} \cdot cyt\ c_{ox} - k_{-33} \cdot cyt\ c1_{ox} \cdot cyt\ c_{red}$$

Conservation relations

$$FeS_{red} = c3_{tot} - FeS_{ox}$$

$$cyt\ c1_{red} = c3_{tot} - cyt\ c1_{ox}$$

$$cyt\ c_{red} = cyt\ c_{tot} - cyt\ c_{ox}$$

TABLE S2. Parameters used in the complexes II and III models

Parameter	Description	Value	Reference
Vsucc	Maximum velocity of SDH	250 mM/min	fit using data from (26)
Km_succ	Michaelis constant for SDH	0.6	(25)
k03	Reverse rate constant for reaction 3	$9.9998 \cdot 10^4$	fit using data from (26)
Keq3	Equilibrium constant for reaction 3	0.6877	(25)
k04	Reverse rate constant for reaction 3	$3.6402 \cdot 10^3$	fit using data from (26)
Keq4_bHox	Equilibrium constant for reaction 4 (b_H oxidized)	129.9853	fit using data from (26)
Keq4_bHred	Equilibrium constant for reaction 4 (b_H reduced)	13.7484	fit using data from (26)
δ_1	Fraction of $\Delta\Psi$ at energy barrier peak for reaction 4	0.5	(25)
α	Fraction of $\Delta\Psi$ affecting Q_p^- to $b_L e^-$ transfer	0.2497	Derived from β value, assuming membrane symmetry
k_d	Rate of diffusion across the membrane for Q and QH_2	$1.32 \cdot 10^6$ /min	(25)
k06	Reverse rate constant for reaction 6	10,000	fit using data from (26)
Keq6	Equilibrium constant for reaction 6	9.4546	fit using data from (26)
δ_2	Fraction of $\Delta\Psi$ at energy barrier peak for reaction 6	0.5	(25)
β	Fraction of $\Delta\Psi$ affecting b_L to $b_H e^-$ transfer	0.5006	fit using data from (26)
k07_bLox	Reverse rate constant for reaction 7 (b_L oxidized)	800	fit using data from (26)
Keq7_bLox	Equilibrium constant for reaction 7 (b_L oxidized)	3.0748	(25)
k07_bLred	Reverse rate constant for reaction 7 (b_L reduced)	100	fit using data from (26)

Keq7_bLred	Equilibrium constant for reaction 7 (b_L reduced)	29.0714	Adjusted from (25) by for anticooperativity as given in (26)
δ_3	Fraction of $\Delta\Psi$ at energy barrier peak for reaction 7	0.5	(25)
γ	Fraction of $\Delta\Psi$ affecting b_H to $Q_n e^-$ transfer	0.2497	Derived from β value, assuming membrane symmetry
k08_bLox	Reverse rate constant for reaction 8 (b_L oxidized)	5000	fit using data from (26)
Keq8_bLox	Equilibrium constant for reaction 8 (b_L oxidized)	129.9853	(26)
k08_bLred	Reverse rate constant for reaction 8 (b_L oxidized)	500	fit using data from (26)
Keq8_bLred	Equilibrium constant for reaction 8 (b_L reduced)	9.4546	(26)
k09	Reverse rate constant for reaction 9	$4.9949 \cdot 10^4$	fit using data from (26)
Keq9	Equilibrium constant for reaction 9	0.2697	(25)
k010	Reverse rate constant for reaction 10	50	fit using data from (26)
Keq10	Equilibrium constant for reaction 10	1.4541	(26)
k33		148,148	(29)
Keq33	Equilibrium constant for reaction 33	2.1145	(26)
C3tot	Total complex III protein	0.325mM	(29)

Equilibrium constants were adjusted to 37°C

Section 3. Cytochrome c Oxidase (Complex IV)

The complex IV model used here is derived from the representation in Demin et al.(29) as depicted in Fig. 1 *D* of the main text. Model parameters were refit to match data on $\Delta\Psi_m$ -dependence of oxygen consumption (5) and cytochrome c redox state (26, 27) The binding affinity for oxygen was set according to measurements by Brzezinski and Adelroth (30). KA Pattern (21) was also applied to this model to derive an algebraic expression for oxygen consumption.

Rate constants of complex II, III and IV reactions were optimized to fit the data shown in Figs. 2-4 from ref. (26) while maintaining the equilibrium constants as determined by redox potentials from the literature. Complex I rate constants were optimized to fit data from ref. (22), as presented in Fig. 7. Experimental data from cardiac mitochondria were used whenever possible.

Rate equations used in complex IV model

$$a_{12} = k_{34} \cdot \exp(-d_5 \cdot 4 \cdot \Delta\Psi \frac{F}{RT}) \cdot \text{cytc}_{\text{red}}^3 \cdot H_n^4$$

$$a_{14} = k_{-37} \cdot \exp((1 - d_5) \cdot \Delta\Psi \frac{F}{RT}) \cdot H_p$$

$$a_{21} = k_{-34} \cdot \exp((1 - d_5) \cdot 4 \cdot \Delta\Psi \frac{F}{RT}) \cdot \text{cytc}_{\text{ox}}^3 \cdot H_p$$

$$a_{23} = k_{35} \cdot O_2 \cdot (1 - \text{CNBlock})$$

$$a_{34} = k_{36} \cdot \exp(-d_5 \cdot 3 \cdot \Delta\Psi \frac{F}{RT}) \cdot \text{cytc}_{\text{red}} \cdot H_n^3$$

$$a_{41} = k_{37} \cdot \exp(-d_5 \cdot \Delta\Psi \frac{F}{RT}) \cdot H_n$$

$$a_{43} = k_{-36} \cdot \exp((1 - d_5) \cdot 3 \cdot \Delta\Psi \frac{F}{RT}) \cdot \text{cytc}_{\text{ox}} \cdot H_p^2$$

$$E_1 = a_{21} \cdot a_{41} \cdot a_{34} + a_{41} \cdot a_{34} \cdot a_{23}$$

$$E_2 = a_{12} \cdot a_{41} \cdot a_{34}$$

$$E_3 = a_{23} \cdot a_{12} \cdot a_{41} + a_{43} \cdot a_{14} \cdot a_{21} + a_{23} \cdot a_{43} \cdot a_{12} + a_{23} \cdot a_{43} \cdot a_{14}$$

$$E_4 = a_{14} \cdot a_{34} \cdot a_{21} + a_{34} \cdot a_{23} \cdot a_{12} + a_{14} \cdot a_{34} \cdot a_{23}$$

$$D = E_1 + E_2 + E_3 + E_4$$

$$Y = E_1/D$$

$$Y_r = E_2/D$$

$$Y_O = E_3/D$$

$$Y_{OH} = E_4/D$$

$$v_{34} = C4_{\text{tot}} \cdot (Y \cdot a_{12} - Y_r \cdot a_{21})$$

$$v_{35} = C4_{\text{tot}} \cdot Y_r \cdot a_{23}$$

$$v_{36} = C4_{\text{tot}} \cdot (a_{34} \cdot Y_O - a_{43} \cdot Y_{OH})$$

$$v_{37} = C4_{\text{tot}} \cdot (a_{41} \cdot Y_{OH} - a_{14} \cdot Y)$$

$$V_e = 3v_{34} + v_{35}$$

$$V_H = 4v_{34} + 3v_{36} + v_{37}$$

$$V_{O_2} = v_{35}$$

$$V_{hRes} = 2v_3 + V_H$$

TABLE S3. Parameters used in complex IV model

Parameter	Description	Value	Reference
cyt c _{tot}	Total cytochrome c	0.325mM	(29)
C4 _{tot}	Total complex IV protein	0.325mM	(29)
k ₃₄		$1.7667 \cdot 10^{28}$	Adjusted from (29) to fit (26)
k ₋₃₄		1.7402	Adjusted from (29) to fit (26)
k ₃₅		45,000	(30)
k ₃₆		$2.8955 \cdot 10^{25}$	Adjusted from (29) to fit (26)
k ₋₃₆		$2.8955 \cdot 10^{10}$	Adjusted from (29) to fit (26)
k ₃₇		$1.7542 \cdot 10^{12}$	Adjusted from (29) to fit (26)
k ₋₃₇		$1.7542 \cdot 10^4$	Adjusted from (29) to fit (26)
δ ₅		0.5	(29)
O ₂	Matrix oxygen concentration	$6 \cdot 10^{-3}$ mM	(29)
O ₂ ⁻	Matrix superoxide concentration	$0.1 \cdot 10^{-3}$ mM	(29)

Section 4. ATP generation and proton leak systems

$$\Delta pH = pH_p - pH_n$$

$$\mu_H = -2.303 \cdot RT/F \cdot \Delta pH + \Delta \Psi$$

$$V_{AF1} = \frac{K_{ATPase_app} \cdot ATP}{P_i \cdot (ADP^{3+} + HADP)}$$

$$V_{hu} = -\rho_{F1} \frac{300 \cdot p_a + 300 \cdot p_a \cdot V_{AF1} - 3e^{\frac{3F}{RT}\mu_H} (p_a + p_b)}{e^{\frac{150F}{RT}} + p_1 \cdot e^{\frac{150F}{RT}} \cdot V_{AF1} + e^{\frac{3F}{RT}\mu_H} (p_2 + p_3 \cdot V_{AF1})}$$

$$V_{h leak} = g_H \cdot \mu_H$$

TABLE S4. Parameters for ATP generation and proton leak systems

Parameter	Description	Value	Reference
C _{mito}	mitochondrial membrane capacitance	1.812·10 ⁻³ mM/mV	(1)
K _{ATPase_app}	Apparent equilibrium constant for ATPase	4.1175· 10 ⁶	
Pi	total concentration of inorganic phosphate	8.6512mM	
HADP	concentration of proton-bound ATP	0.0026mM	
ρ _{F1}	concentration of ATPase	5mM	
p _a		9.936·10 ⁻⁴ min ⁻¹	(4)
p _b		2.0238·10 ⁻⁵ min ⁻¹	(4)
p ₁		1.346·10 ⁻⁴	(4)
p ₂		7.739·10 ⁻⁷	(4)
p ₃		6.65·10 ⁻¹⁵	(4)

Section 5. Simulation Settings

Inputs to the model (with the exception of Fig.10) are cytosolic pH (pH_p), matrix pH (pH_n), $\Delta\Psi$ (if clamped), NADH, NAD, percent maximal succinate dehydrogenase activity, myxothiazol inhibition, antimycin inhibition, rotenone inhibition, cyanide inhibition, mitochondrial membrane proton leak conductance (gh), matrix ATP, and matrix ADP. Fig. 10 presents results from the isolated complex I model using clamped $\Delta\Psi$ and Q and QH_2 as additional inputs. The values of these inputs for each figure are given in the tables below. Inhibitors are off unless otherwise stated. In the absence of succinate or malate measurements, SDH activity was fit using the data.

TABLE S5. Simulation parameters for Figs. 2A-C

Parameter	Value
pH_p	7.0
pH_n	7.3
$\Delta\Psi$	clamped to values ranging from 70 to 200mV
NADH	1mM
NAD^+	9mM
SDH activity	60%
gh	0.12mM/min/mV
ATP	0.01mM
ADP	0.0110mM

NADH:NAD ratio comes from most reduced estimate by Mintz and Robin (31) for alveolar macrophages to compare to Kim et al. (26) RAW macrophages. SDH activity was increased to fit the Q/ QH_2 redox data in accordance with the observation from a previously published mitochondria model (13) that SDH activity is increased as NADH becomes oxidized.

TABLE S6. Simulation parameters for Figs. 3 and 5 B

Parameter	Value
pH_p	7.0
pH_n	7.6
gh	0.12mM/min/mV
ATP	0.01mM
ADP	0.0110mM
RET-4 $\Delta\Psi$	195mV
RET-4 NADH	8.168mM
RET-4 NAD^+	1.832mM
RET-4 SDH activity	50%
RET-3 $\Delta\Psi$	150mV
RET-3 NADH	2.560mM
RET-3 NAD^+	7.440mM
RET-3 SDH activity	50%

RET-4 $\Delta\Psi$	190mV
FET-4 NADH	3.489mM
FET-4 NAD ⁺	6.511mM
FET-4 SDH activity	5%
FET-3 $\Delta\Psi$	155mV
FET-3 NADH	2.147mM
FET-3 NAD ⁺	7.853mM
FET-3 SDH activity	5%

$\Delta\Psi$ values from Aon et al. (22), their figures 3B and C. NAD⁺/NADH estimates from Muraoka and Slater (23) 10mM pyruvate protocol. SDH activity for FET simulations was kept low to reproduce the trend of increasingly oxidized Q for RET-4, RET-3, FET-4, followed by FET-3 shown in the pyruvate and succinate data from Muraoka and Slater (23).

TABLE S7. Simulation parameters for Figs. 4 and S3 B

Parameter	Value
pH _p	7.0
pH _n	7.6
$\Delta\Psi$	clamped to values ranging from 0 to 200mV
NADH	2.3055mM
NAD ⁺	7.685mM
SDH activity	100%
gh	0.12mM/min/mV
ATP	0.01mM
ADP	0.0110mM
rotenone block	100%

TABLE S8. Simulation parameters for Fig. 5 A

Parameter	Value
pH _p	7.2
pH _n	7.3 to 7.8
$\Delta\Psi$	200mV
NADH	2.5mM
NAD ⁺	7.5mM
SDH activity	100%
gh	0.12mM/min/mV
ATP	0.01mM
ADP	0.0110mM

Fig. 5 B uses conditions from Fig. 3 with pH_n = {7.6, 8.0, 8.3}

TABLE S9. Simulation parameters for Fig. 6 A

Parameter	Value
pH _p	7.0
pH _n	7.6
gh	0.12mM/min/mV

ATP	0.01mM
ADP	0.0110mM
G/M	
NADH	3.489mM
NAD ⁺	6.511mM
$\Delta\Psi$	unclamped
SDH activity	20%
G/M + rot	
NADH	3.489mM
NAD ⁺	6.511mM
$\Delta\Psi$	unclamped
SDH activity	20%
rotenone block	95%

NAD⁺/NADH estimates from Muraoka and Slater (23) 10mM pyruvate protocol.

TABLE S10. Simulation parameters for Fig. 6 B

Parameter	Value
pH _p	7.0
pH _n	7.6
gh	0.12mM/min/mV
ATP	0.01mM
ADP	0.0110mM
succ + rot	
NADH	0mM
NAD ⁺	10mM
$\Delta\Psi$	unclamped
SDH activity	50%
rotenone block	95%
succ + rot + AA	
NADH	0mM
NAD ⁺	10mM
$\Delta\Psi$	unclamped
SDH activity	50%
rotenone block	95%
antimycin A block	77.5%
succ + rot + myx	
NADH	0mM
NAD ⁺	10mM
$\Delta\Psi$	unclamped
SDH activity	50%
rotenone block	95%
myxothiazol block	85%

TABLE S11. Simulation parameters for Fig. 7

Parameter	Value
pH _p	7.0
pH _n	7.6
$\Delta\Psi$	unclamped
NADH	3mM
NAD ⁺	7mM
SDH activity	20%
gh	varied from 0.12 to 8.4 mM/min/mV
ATP	0.01mM
ADP	0.0110mM

TABLE S12. Simulation parameters for Fig. S1

Parameter	Value
pH _p	7.0
pH _n	7.5
$\Delta\Psi$	129.2445mV
NADH	0.03mM
NAD ⁺	ranges from 10 ⁻¹⁰ to 10 ⁻⁵ mM
Q	0mM
QH ₂	0mM
rotenone block	100%

TABLE S13. Simulation parameters for Fig. S2

Parameter	Value
pH _p	7.0
pH _n	7.1, 7.3, 7.9
$\Delta\Psi$	200mV
NADH	3.489mM
NAD ⁺	6.511mM
SDH activity	5%
gh	0.12mM/min/mV
ATP	0.01mM
ADP	0.0110mM

TABLE S14. Simulation parameters for Fig. S3 A

Parameter	Value
pH _p	7.0
pH _n	7.6
$\Delta\Psi$	clamped to values ranging from 0 to 200mV
NADH	2.3055mM
NAD ⁺	7.685mM

SDH activity	20%
gh	0.12mM/min/mV
ATP	0.01mM
ADP	0.0110mM

Fig. S3B uses the same settings as Fig. 4

TABLE S15. Simulation parameters for Fig. S4

Parameter	Value
pH _p	7.0
pH _n	7.3
$\Delta\Psi$	unclamped
SDH activity	20%
gh	0.12mM/min/mV
NADH	Wei et al. recordings, their figure 4D
NAD ⁺	10-NADH
state 0 ATP	0.01mM
state 0 ADP	0.0110mM
state 4 ATP	1.2223mM
state 4 ADP	0.0110mM
state 3 ATP	1.2223mM
state 3 ADP	1.0mM

Section 6. Minimal scavenging model

Equations and parameters are reproduced from the code for simulations in Aon et al. (2).

System of differential equations describing ROS scavenging

$$d[H_2O_2]/dt = V_{SOD} - k_1 \cdot [H_2O_2] - V_{Prx} - V_{GPx}$$

$$d[GSH]/dt = V_{GR} - V_{GPx}$$

$$d[Trx(SH)_2]/dt = V_{TrxR} - V_{TxPx}$$

$$d[Prx(SH)_2]/dt = V_{TxPx} - V_{Prx}$$

Rate equations for ROS scavenging

Rate of glutathione peroxidase (GPx), oxidizes GSH, detoxifies H_2O_2

$$V_{GPx} = \frac{E_T^{GPx} [H_2O_2] [GSH]}{\Phi_1 [GSH] + \Phi_2 [H_2O_2]}$$

Rate of glutathione reductase (GR), reduces GSSG to 2GSH

$$V_{GR} = \frac{k_{GR} E_T^{GR} [GSSG] [NADPH]}{[GSSG] [NADPH] + K_M^{GSSG} [NADPH] + K_M^{NADPH} [GSSG] + K_M^{GSSG} K_M^{NADPH}}$$

Rate of peroxiredoxin (a.k.a. thioredoxin-dependent peroxidase) oxidation, detoxifies H_2O_2

$$V_{Prx} = \frac{E_T^{Prx} [H_2O_2] [Prx(SH)_2]^2}{\Phi_{P2} [Prx(SH)_2] + \Phi_{P1} [H_2O_2]}$$

Rate of thioredoxin reductase (TrxR)

$$V_{TrxR} = \frac{k_{TrxR} E_T^{TrxR} [Trx(SS)] [NADPH]}{[Trx(SS)] [NADPH] + K_M^{Trx(SS)} [NADPH] + K_M^{T-NADPH} [Trx(SS)] + K_M^{Trx(SS)} K_M^{T-NADPH}}$$

Rate of peroxiredoxin reduction and thioredoxin oxidation

$$V_{TxPx} = k_{TxPx} [Trx(SH)_2] [Prx(SS)]$$

Conservation relations

$$GSSG = 0.5 \cdot (G_{tot} - GSH)$$

$$Trx(SS) = Trx_{tot} - Trx(SH)_2$$

$$Prx(SS) = Prx_{tot} - Prx(SH)_2$$

TABLE S16. Parameters for scavenging model

Parameter	Description	Value
E_T^{GPx}	concentration of GPx	0.0034mM
Φ_1	GPx activity constant	$1.1337 \cdot 10^{-7}/\text{min}$
Φ_2	GPx activity constant	$2.5 \cdot 10^{-5}/\text{min}$
E_T^{GR}	concentration of GR	$5 \cdot 10^{-3}\text{mM}$
k_{GR}	rate constant of GR	1500/min

K_M^{GSSG}	K_M of GR for oxidized GSH	0.12mM
K_M^{NADPH}	K_M of GR for NADPH	0.015mM
G_{tot}	total glutathione pool	3mM
E_T^{Prx}	concentration of Prx	1mM
Φ_{P1}	Prx activity constant	$6.3833 \cdot 10^{-6}/\text{min}$
Φ_{P2}	Prx activity constant	$3.0833 \cdot 10^{-7}/\text{min}$
$K_M^{Trx(SS)}$	K_M of TrxR for Trx(SS)	0.006
$K_M^{T-NADPH}$	K_M of TrxR for NADPH	0.012
k_{TrxR}	rate constant of TrxR	$1.32 \cdot 10^3/\text{min}$
E_T^{TrxR}	concentration of TrxR	0.01
k_{TxPx}	rate constant of Prx	$1.5 \cdot 10^5/\text{min}$
Trx_{tot}	total thioredoxin pool	0.025
Prx_{tot}	total peroxiredoxin pool	0.15
k_1	rate constant of H_2O_2 transport from mitochondria	0.002

Section 7. Initial conditions

TABLE S17. Initial conditions

State variable	Value
Q_n	1.2518 mM
Q_n^-	0.5578 mM
$(QH_2)_n$	0.4199 mM
$(QH_2)_p$	0.4197 mM
Q_p^-	0.0988 mM
Q_p	1.2520 mM
b1	0.0618 mM
b2	0.1500 mM
b3	0.0954 mM
b4	0.0179 mM
FeS_{ox}	0.0900 mM
FeS_{red}	0.2350 mM
cyt c1 _{ox}	0.2266 mM
cyt c _{ox}	0.1785 mM
cyt c1 _{red}	0.0984 mM
cyt c _{red}	0.1465 mM
$\Delta\Psi$ (only relevant if unclamped)	180mV
H ₂ O ₂ (only with scavenging model)	0 mM
GSH (only with scavenging model)	2.8 mM
Trx(SH) ₂ (only with scavenging model)	0.002 mM
Prx(SH) ₂ (only with scavenging model)	0.0281 mM

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